

Home SOC Suite — Red Team Whitepaper

Executive Summary

This whitepaper presents a red team analysis of the Home SOC Suite, a modular host-based monitoring and detection system. The analysis identifies critical weaknesses in baseline detection models, correlation strategies, and data integrity assumptions. It further defines architectural improvements that transform the system into a resilient behavioral detection platform. The findings demonstrate that effective detection is not solely dependent on data collection, but on the ability to reason about time, trust, and contextual consistency across multiple telemetry sources.

System Architecture Overview

The system is composed of three primary layers: 1. Collection Layer (PowerShell): Gathers telemetry across system, network, persistence, and process activity. 2. Correlation Layer (Python): Normalizes data, applies detection rules, and computes behavioral risk scores. 3. Analysis & Forensic Layer: Provides dashboard visibility and archive-based investigation capabilities. This layered approach mirrors real-world SOC pipelines and enables modular expansion.

Threat Model

The system assumes a local attacker with user or administrator-level access. Kernel-level compromise is considered out of scope. Adversary goals include: - Establishing persistence - Maintaining command and control - Evading detection - Manipulating or destroying forensic evidence

Attack Scenario Summary

Attack	Technique	Impact	Mitigation
Baseline Poisoning	Behavior normalization abuse	Stealth persistence	Baseline versioning & drift detection
Time Desync	Split execution	Missed correlation	Multi-timescale correlation
Trusted Binary Abuse	Living-off-the-land	Stealth execution	Context-aware process analysis
Noise Flooding	Signal masking	Alert fatigue	Rate anomaly detection
Collector Suppression	Telemetry removal	Blind spots	Heartbeat monitoring
Low-and-Slow C2	Rare communication	Evasion	Rarity scoring
Archive Tampering	Log modification	Forensic compromise	Signed manifests
Log Injection	Parser confusion	False data	Schema validation
Replay Attacks	Event reordering	Broken timelines	Timestamp separation
Cross-Source Drift	Inconsistency	False trust	Cross-validation
Identity Abuse	Context shifting	Hidden intent	User/session tracking

Detection Model Design

The detection engine combines rule-based logic with a behavioral scoring model. Rules detect explicit malicious patterns, while scoring aggregates weak signals such as rarity, timing anomalies, and baseline deviation. Alerts are generated when thresholds or rule conditions are met, enabling both explainability and flexibility.

Key Architectural Improvements

The hardened system introduces the following capabilities: - Multi-timescale correlation (minutes, hours, days) - Trust-aware event modeling - Collector health monitoring and missing data detection - Baseline lifecycle management - Archive integrity validation with cryptographic chaining - Cross-collector validation for consistency checks - Identity-aware detection logic These improvements significantly increase resilience against evasion techniques.

System Comparison

Capability	Baseline System	Hardened System
Correlation	Short window only	Multi-timescale
Baseline	Static	Versioned + adaptive
Integrity	Encryption only	Signed + chained
Detection	Event-based	Behavior-based
Visibility	No health checks	Full telemetry monitoring

Conclusion

The Home SOC Suite evolves from a monitoring tool into a behavioral detection system through architectural refinement. The introduction of time-awareness, trust modeling, and integrity validation transforms it into a robust defensive platform. The primary takeaway is that security effectiveness is determined not by visibility alone, but by the system's ability to interpret and validate that visibility under adversarial conditions.